Pratheek Eranki

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Summary

Software engineer with experience in enterprise application delivery, sleep EEG and fMRI research data preparation, and legislative data analysis. Builds full-stack applications and AI/data projects using Python, Java, SQL, React, and machine learning tools. Co-author of the Rhamba research publication.

Education

University of Memphis Aug 2024 – May 2026
M.S. in Data Science GPA 3.87 / 4.0

Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, R, Scala

Deep learning & architectures: PyTorch, TensorFlow, Transformers / self-attention, Mamba / state space models, Self-supervised learning, Masked autoencoders, Model fine-tuning, CBraMod

LLMs & retrieval: Hugging Face Transformers, Sentence Transformers, RAG, FAISS, LangChain, NVIDIA NIM, LLM APIs (OpenAI, Anthropic, Gemini), Prompt engineering

Machine learning & evaluation: scikit-learn, XGBoost, LightGBM, Time-series forecasting, Cross-validation, SHAP, BLEU / ROUGE / F1 evaluation

Research & ML compute: Resting-state fMRI, EEG sleep staging, MNE-Python, CUDA, Multi-GPU training, HPC clusters

Data & visualization: pandas, NumPy, SciPy, ETL pipelines, Apache Spark, Tableau, Plotly, Matplotlib, Seaborn, Excel

Backend & APIs: Spring Boot, Spring Security, FastAPI, REST APIs, JWT authentication, WebSockets, Apache Kafka

Frontend & applications: React, HTML, CSS, Tailwind CSS, Streamlit, Gradio

Databases & persistence: PostgreSQL, MySQL, MongoDB, Redis, SQLAlchemy, Spring Data JPA, Relational database design, Query optimization

MLOps, delivery & practices: MLflow, Evidently, AWS, Docker, Docker Compose, Git, GitHub Actions / CI/CD, JUnit, pytest, Agile / Scrum, Technical documentation

Experience

St. Jude Children's Research Hospital September 2025 – May 2026
Software Engineer

Contributed to software and data workflows supporting sleep EEG and fMRI research in a clinical research environment.

- Prepared and validated sleep EEG and fMRI data for downstream research analysis, resolving data-quality and formatting issues while maintaining traceable processing steps.
- Used PyTorch and TensorFlow in research workflows involving Transformers, self-attention, and Mamba architectures; supported model preparation, training, and fine-tuning experiments.
- Worked with CBraMod-based EEG sleep-staging workflows and self-supervised fMRI experiments, connecting data preparation with model inputs and downstream evaluation.
- Collaborated with researchers to translate study requirements into repeatable processing and analysis workflows, and developed utilities to reduce repetitive preparation work.
- Documented processing procedures, validation checks, experiment configurations, and technical decisions so research workflows could be reproduced, reviewed, and maintained.

University of Memphis March 2025 – August 2025
Graduate Student Research Assistant

Conducted data-focused research using legislative records obtained through LegiScan.

- Built reproducible workflows to collect legislative records through the LegiScan API, clean source data, and organize structured SQL datasets for research analysis.
- Standardized bill identifiers, jurisdictions, legislative status, sponsors, and subject classifications to support consistent comparisons across legislative records.
- Used SQL and Excel to clean, deduplicate, and align legislative records with demographic data by state and year for comparative analysis.
- Performed exploratory analysis of legislative activity and prepared Tableau and Plotly views to communicate trends and structured findings for faculty review.
- Documented data definitions, transformation logic, API collection steps, and known limitations so research workflows and datasets could be reproduced and extended.

Cognizant October 2021 – April 2024
Software Engineer

Contributed to enterprise software delivery in client-facing engineering environments.

- Developed and maintained Java Spring Boot application components and REST APIs within established development, testing, and release processes.
- Built and maintained React and TypeScript interface components for data display, form validation, and application workflows, integrating them with backend services.
- Worked with MySQL queries and indexing to support application data access and reporting; investigated software defects, documented root causes, and coordinated fixes with engineering and business stakeholders.
- Supported JUnit testing, Docker-based environments, AWS deployment workflows, CI/CD, release validation, and production issue resolution across iterative software deliveries.
- Participated in Agile delivery and code reviews, and created technical documentation and handoff materials explaining application behavior, recurring issues, and maintenance procedures.

Projects

FieldVerify Code | Case study

FastAPI / React / Android / Identity SDKs

- Built the workflow around separate authenticity and matching signals, manual fallback, SDK integration, and a chained audit log.
- Uses a FastAPI backend with React and Android interfaces.
- Separates cryptographic credential checks from record-match scoring.
- Records outcomes in a SHA-256-chained audit trail and includes tests for matching, audit behavior, and verification paths.

OpsFlow Code | Case study

Spring Boot / React / PostgreSQL / WebSockets

- Implemented team and ticket workflows across Spring Boot services and the React interface, including role-aware access and real-time updates.
- Scopes organizations, users, projects, and tickets through authenticated APIs.
- Uses WebSockets for board updates and Kafka for activity events.
- Combines PostgreSQL persistence, Redis caching, and container-based development.

FixBot Code | Case study

Python / NVIDIA NIM / FAISS / FastAPI / Gradio

- Implemented a custom iterative tool-calling loop, structured lookup tools, session memory, FastAPI endpoints, and a Gradio interface showing tool activity.
- Dispatches three tools for symptoms, pricing, and FAISS-backed knowledge search.
- Feeds tool results back into an OpenAI-compatible NVIDIA NIM model until a response is ready.
- Keeps a sliding window of conversation history and exposes the API separately from the interface.

Publications

Pankaj Pandey, Ruthwik Reddy Doodipala, Pratheek Eranki, Carolina Torres-Rojas, Manob Jyoti Saikia, Ranganatha Sitaram. *Rhamba: Region-Aware Hybrid Attention-Mamba Framework for Self-Supervised Learning in Resting-State fMRI*. Accepted for publication in Computers in Biology and Medicine, as reported by the arXiv record. arXiv:2605.01240.

Certifications

- AWS Certified Cloud Practitioner — Amazon Web Services Nov 2023 · Expires Nov 2026
- Microsoft Certified: Azure Fundamentals — Microsoft Apr 2023
- AI Fluency: Framework & Foundations — Anthropic Jun 2026
- Crash Course on Python — Coursera Jun 2020

Credentials on LinkedIn